The exponential moon is the missing winding-number counterexample

Full negative answer to an open question in arXiv:2406.07656

17 August 2026

Status. Candidate full counterexample, together with a sharp classification of all symbols e^{az}. This is a new-result claim subject to human review.

Source question

González and León-Saavedra prove that constant winding number is necessary for an entire symbol to induce an analytic Toeplitz operator with the double commutant property. They ask whether it is sufficient, and specifically suggest $\varphi(z) = e^{\pi i z}$ as a possible counterexample.

> class (see in [17, p. 55] this question formulated). A question that awaits to be answered is characterizing the elements of the disk algebra that induce multiplication operators with the double commutant property.
>
> The necessary condition on the winding number is somewhat unexpected. We would like to know whether it is sufficient or not for the case of entire functions. We suspect that it is not a sufficient condition. A good counterexample is to find an entire function that takes the disk to a moon-shaped region such that the kissing point has very strong contact (see [16, p. 22]). Professor Alicia Cantón Pire suggested a good starting point for this question by considering the exponential function $\exp(\pi i z)$. This function is injective on the unit disk and takes the unit disk to a moon-shaped region with a very strong contact in the kissing point.

Here M_φ denotes multiplication by φ on $H^2(\mathbb{D})$, and the double commutant property is

$$\overline{\mathrm{Alg}(M_\varphi)}^{\mathrm{WOT}} = \{M_\varphi\}''.$$

Main result

Theorem 1. *The entire function*

$$\varphi(z) = e^{\pi i z}$$

has winding number one about every point of $\varphi(\mathbb{D})$, but M_φ does not have the double commutant property. Consequently, the winding-number condition in arXiv:2406.07656 is not sufficient.

More generally, for $a \in \mathbb{C}$,

$$M_{e^{az}} \text{ has the double commutant property} \iff a = 0 \text{ or } |a| < \pi.$$

Exact geometry of the critical image

Let $\Omega = \varphi(\mathbb{D})$. If $\varphi(z_1) = \varphi(z_2)$, then $z_1 - z_2 = 2k$ for some $k \in \mathbb{Z}$. Since two points of $\mathbb{D}$ have distance strictly less than two, φ is univalent on $\mathbb{D}$.

Write $w = re^{i\theta}$ with $-\pi < \theta < \pi$. The inverse branch is

$$z = \frac{\theta}{\pi} - \frac{i}{\pi} \log r,$$

so

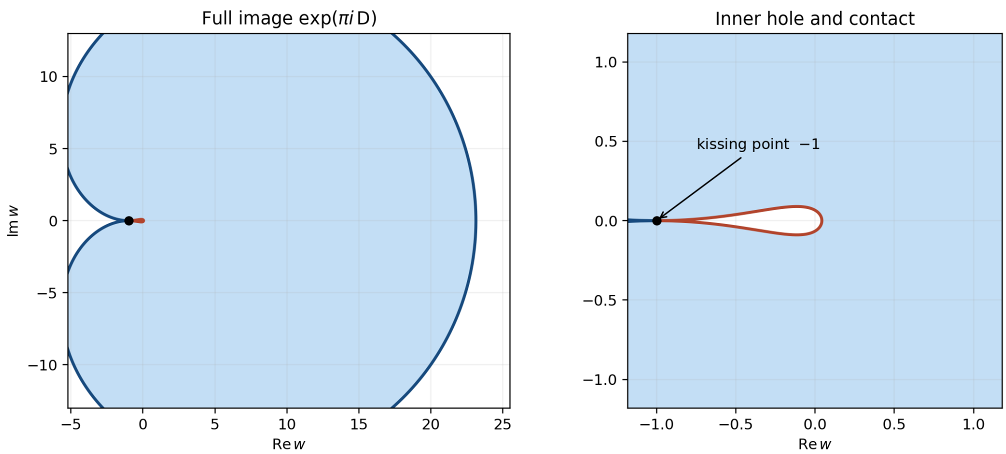
$$\Omega = \left\{ re^{i\theta} : -\pi < \theta < \pi, \quad \theta^2 + (\log r)^2 < \pi^2 \right\}. \tag{1}$$

Its boundary is the union of two Jordan loops

$$\Gamma_\pm : \qquad r = r_\pm(\theta) := \exp\left(\pm\sqrt{\pi^2 - \theta^2} \right), \qquad -\pi \le \theta \le \pi.$$

They meet only at -1 (the two endpoint arguments $\pm\pi$). Let B be the Jordan domain inside the outer loop Γ_+ and let H be the Jordan domain inside the inner loop Γ_-. Equation (1) gives

$$\Omega = B \setminus \overline{H}, \qquad \overline{H} \cap \partial B = \{-1\}. \tag{2}$$



The relative-hull obstruction

The subtlety is that failure of bounded polynomial approximation by a single sequence does not by itself exclude weak-star generation: Sarason's closure process may have transfinite order. The following argument uses his full criterion.

Recall that a crosscut of a simply connected domain is a Jordan arc whose open arc lies in the domain and whose two distinct endpoints lie on its boundary. We claim that every crosscut of B meets Ω. Indeed, if a crosscut J missed Ω, then by (2)

$$J^\circ \subseteq \overline{H}.$$

Taking closures would put both endpoints of J in $\overline{H} \cap \partial B = \{-1\}$, contradicting their distinctness.

Sarason's Proposition 4 and its corollary now imply that the relative B-hull of Ω is all of B. By the analytic definition of the relative hull, every $f \in H^\infty(B)$ satisfies

$$\sup_B |f| = \sup_\Omega |f|. \tag{3}$$

Since B properly contains Ω, Sarason's Corollary 2 says that a Riemann map from $\mathbb{D}$ onto Ω is not a weak-star generator of $H^\infty(\mathbb{D})$. Thus φ is not a weak-star generator. Corollary 2.8 of the source paper states that for a univalent symbol, the double commutant property is equivalent to weak-star generation. Hence M_φ fails the double commutant property.

On the other hand, φ' never vanishes and every $c \in \Omega$ has exactly one preimage in $\mathbb{D}$. No such c lies on $\varphi(\partial\mathbb{D})$. The argument principle therefore gives

$$n(\varphi(\partial\mathbb{D}), c) = 1 \qquad (c \in \Omega).$$

This proves the counterexample assertion.

Upgrade: the complete exponential phase transition

Put $\varphi_a(z) = e^{az}$, $a \neq 0$, and let

$$p = \frac{2\pi i}{a}$$

be a primitive period. If $|a| < \pi$, then $|p| > 2$; hence φ_a is injective on $\overline{\mathbb{D}}$ and maps the circle to a Jordan curve. Walsh's theorem gives weak-star density of the polynomials in φ_a, and the source criterion gives the double commutant property.

If $|a| = \pi$, the rotation $z \mapsto (a/(\pi i))z$ reduces the image to the critical moon above, so the property fails.

Finally suppose $|a| > \pi$, and put $s = |p| < 2$. Choose a real-plane unit vector $u \perp p$. Select

$$\sqrt{\max\{0, 1 - s^2\}} < r < 1.$$

The value $\varphi_a(ru)$ then has exactly one preimage in $\mathbb{D}$, because all its preimages are $ru + kp$ and $|ru + kp|^2 = r^2 + k^2 s^2 > 1$ for $k \neq 0$. In contrast, choose

$$0 < \varepsilon < \sqrt{1 - s^2/4}$$

avoiding the finitely many values for which $(k - \frac{1}{2})^2 s^2 + \varepsilon^2 = 1$ for some $k \in \mathbb{Z}$, and set $q = -p/2 + \varepsilon u$. Then

$$|q + kp|^2 = (k - \tfrac{1}{2})^2 s^2 + \varepsilon^2.$$

In particular, q and $q + p$ lie in $\mathbb{D}$, while no translate lies on $\partial\mathbb{D}$. Thus $\varphi_a(q)$ has at least two interior preimages and is not on the boundary image. By the argument principle the winding number is nonconstant, and Theorem 5.9 of the source paper rules out the double commutant property. For $a = 0$, the operator is scalar and the property is immediate. This completes the proof of Theorem 1. $\square$

Verification and novelty

The included script `code/verify_exponential_family.py` checks the polar inverse formula on 20,000 random disk points, checks the two boundary arcs and the $|a| > \pi$ preimage witnesses, and numerically computes winding number one for five interior test values using 200,001 boundary samples. It also generates the displayed figure. These checks support, but are not used in place of, the exact proof.

A bounded novelty search covered the run indexes, the exact arXiv id, searches combining $e^{\pi i z}$ with double commutants and weak-star generation, and later citations visible in exact-title searches. No later resolution was found. Novelty confidence is moderate, not exhaustive.

References

1. M. J. González and F. León-Saavedra, *Minimal commutant and double commutant property for analytic Toeplitz operators*, arXiv:2406.07656; especially Corollary 2.8, Theorem 5.9, and Section 6.

2. D. Sarason, *Weak-star generators of H^∞*, Pacific J. Math. **17** (1966), 519–528; especially Proposition 4 and Corollary 2 on pp. 525–527.